

Scenario / Task	What Happens / Why	What to Do / Best Practice	Python Snippet Example
1. Checking Data Info	Understand data size, types, nulls	Use .info() and .describe()	df.info()df.describe()
2. Handling Missing Values	Missing data can bias or break code	Identify missing with .isnull().sum()Fill or drop	df.isnull().sum()df.dropna()df.fillna(value)
3. Identifying & Removing Duplicates	Duplicate rows inflate counts/metrics	Use .duplicated() to find.drop_duplicates() to remove	duplicates = df[df.duplicated()]df_clean = df.drop_duplicates()
4. Analyzing Categorical Variables	Understand category frequency	Use value_counts() or seaborn countplot	df['col'].value_counts()sns.countplot(data=df, x='col')
5. Analyzing Numeric Distributions	See spread, central tendency, skew	Use histograms, boxplotsCheck .skew()	sns.histplot(df['col'])df['col'].skew()
6. Handling Outliers	Extreme values skew mean & analysis	Detect with IQR or Z-scoreRemove or cap	(code snippet you want)
Q1 = df['col'].quantile(0.25) Q3 = df['col'].quantile(0.75) IQR = Q3 - Q1 df_no_outliers = df[(df['col'] >= Q1 - 1.5*IQR) & (df['col'] <= Q3 + 1.5*IQR)]			
7. Calculating Correlations Check linear relationships Use `.corr()` or `scipy.stats` `df[['col1','col2']].corr()` `from scipy.stats import pearsonr`			
8. Grouping & Aggregations Summarize data by categories Use `.groupby()` with aggregation functions `df.groupby('category')['value'].median()`			
9. Creating Bins or Categories Group numeric data into ranges Use `pd.cut()` or `pd.qcut()` `df['binned'] = pd.cut(df['num_col'], bins=[0,5,10,20])`			
10. Encoding Categorical Variables Convert text categories for ML Use one-hot encoding or label encoding `pd.get_dummies(df['cat_col'])`			
11. Handling High Cardinality Too many unique categories Group rare categories into "Other" ```python counts = df['cat_col'].value_counts()			

```
others = counts[counts < threshold].index
```

```
df['cat_col'] = df['cat_col'].replace(others, 'Other')
```

```
| **12. Visualizing Relationships** | Explore trends & patterns | Scatter plots, line plots, bar  
plots, heatmaps | `sns.scatterplot(x='col1', y='col2', data=df)` <br> `sns.heatmap(df.corr())` |
```

```
| **13. Dealing with Text Data in Columns** | Columns with multiple semicolon separated values |  
Split and encode or expand as needed | ```python
```

```
df['skills'] = df['skills'].str.split(';')
```

```
# Or create indicator columns
```

```
from sklearn.preprocessing import MultiLabelBinarizer
```

```
mlb = MultiLabelBinarizer()
```

```
skills_encoded = pd.DataFrame(mlb.fit_transform(df['skills']), columns=mlb.classes_)
```

```
| **14. Data Cleaning: Fixing Data Types** | Numeric stored as strings or inconsistent types |  
Convert types for analysis | `df['col'] = pd.to_numeric(df['col'], errors='coerce')` |
```

```
| **15. Saving Cleaned Data** | Save progress for later use | Export to CSV or Excel  
| `df.to_csv('cleaned_data.csv', index=False)` |
```

-**Missing Data Strategy:**

- If missing <5%, drop rows.
- If important columns missing, consider imputing mean/median or a special category.

-**Outliers Treatment:**

- For salary or income, remove values > 3 std deviations or outside 1.5 IQR.
- Alternatively, use median instead of mean.

-**Duplicates:**

- Remove exact duplicate rows.
- If near-duplicates exist, decide on columns that define uniqueness.

-**Correlation Interpretation:**

- Correlation near 0 = no linear relationship.
- Correlation close to +1/-1 = strong positive/negative linear relationship.